FC6: Parameterized Function for Analogue Signal Acquisition

- FC6 is a routine with **formal input/output parameters** that have to be replaced by real operands when the function is called
- **Input parameters**
  - **ACQ**: 16-bit raw data coming from the analogue board
  - **SS** and **SI**: High and Low thresholds, 32-bit floating point numbers
  - **a** and **b**: conversion parameters, 32-bit floating point numbers
- **Output parameters**
  - **STAT**: status byte, 8 bits
  - **VAL**: result of the conversion, 32-bit floating point number

- FC6 has been **designed for Simatic S7 PLCs**
- The analogue board uses a **16-bit A/D converter**
- In case of **acquisition error**, the raw data from the board is **7FFF**
- The conversion law applied in this case is linear:

  $\boldsymbol{y = ax + b}$
- $\boldsymbol{y}$ is the parameter VAL; $\boldsymbol{x}$ is the parameter ACQ, $\boldsymbol{a}$ and $\boldsymbol{b}$ are the specified input parameters

FC6: Program Flow

- The flow of FC6 is similar to that of FC5, with the following major differences:
  - In case a threshold **is exceeded, FC6 returns the converted signal** (FC5 returned the error code FFFF)
  - The status **byte uses a cumulative error bit**: if there has been an acquisition error, or a threshold has been exeeded, then, besides the corresponding error bit, **also bit #7 is set to 1**. This is commonly used to rapidly check **whether an error occurred**
  - In case of acquisition error, **the function returns the hex constant FFFFFFFF**
- Note: engineering values are expressed with **scientific notation**
  - For example, 3E02 = 300, 2E-01 = 0.2